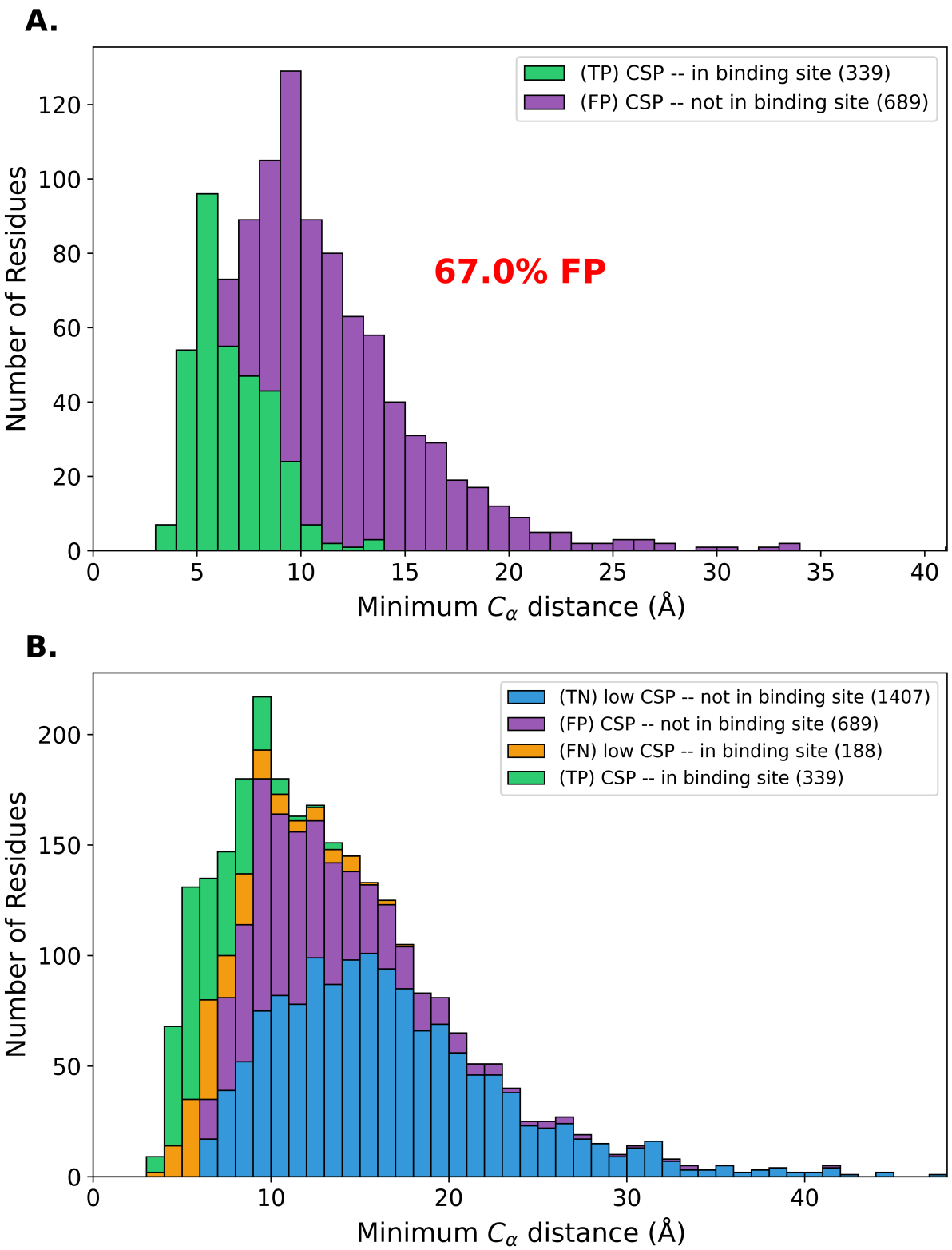
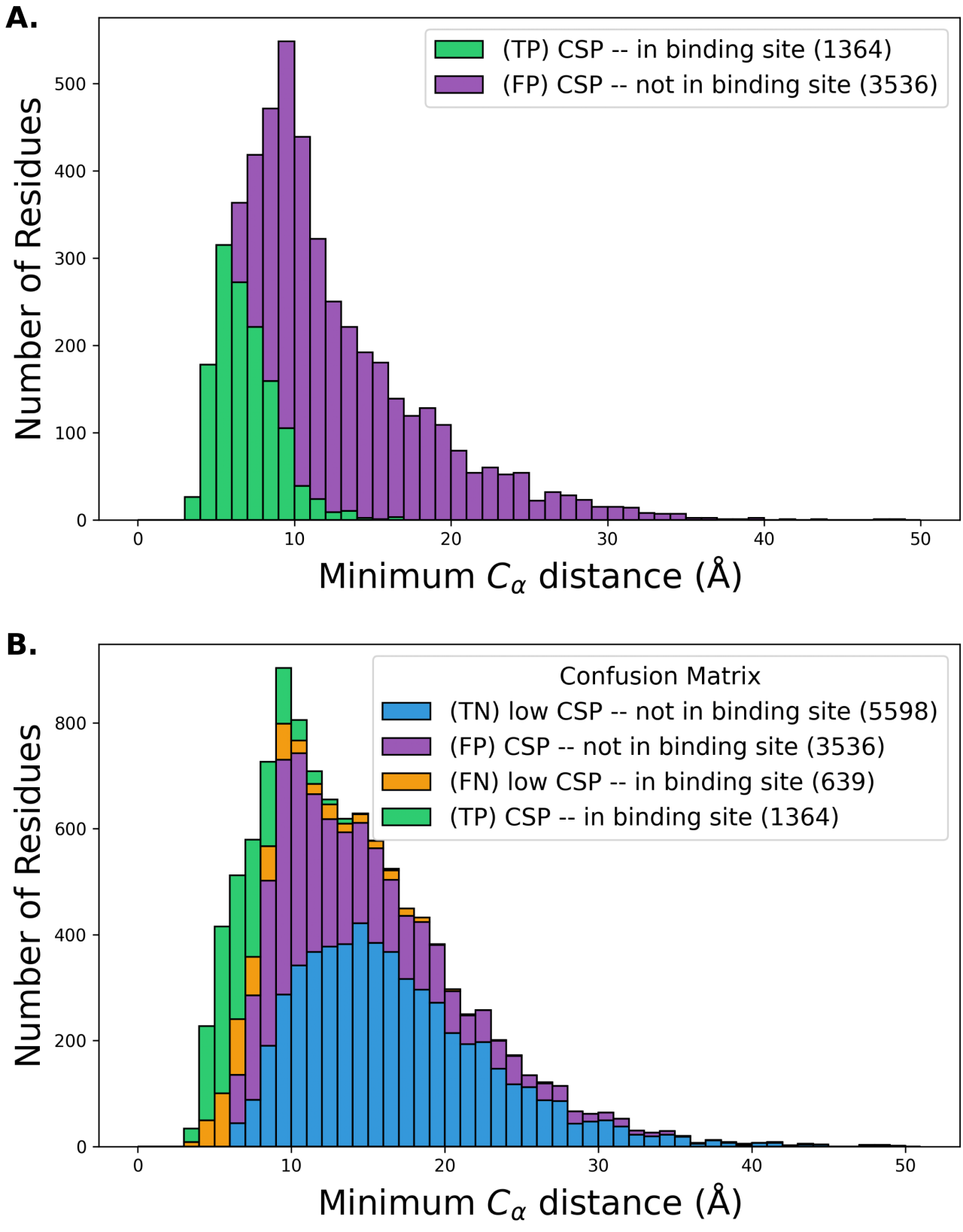
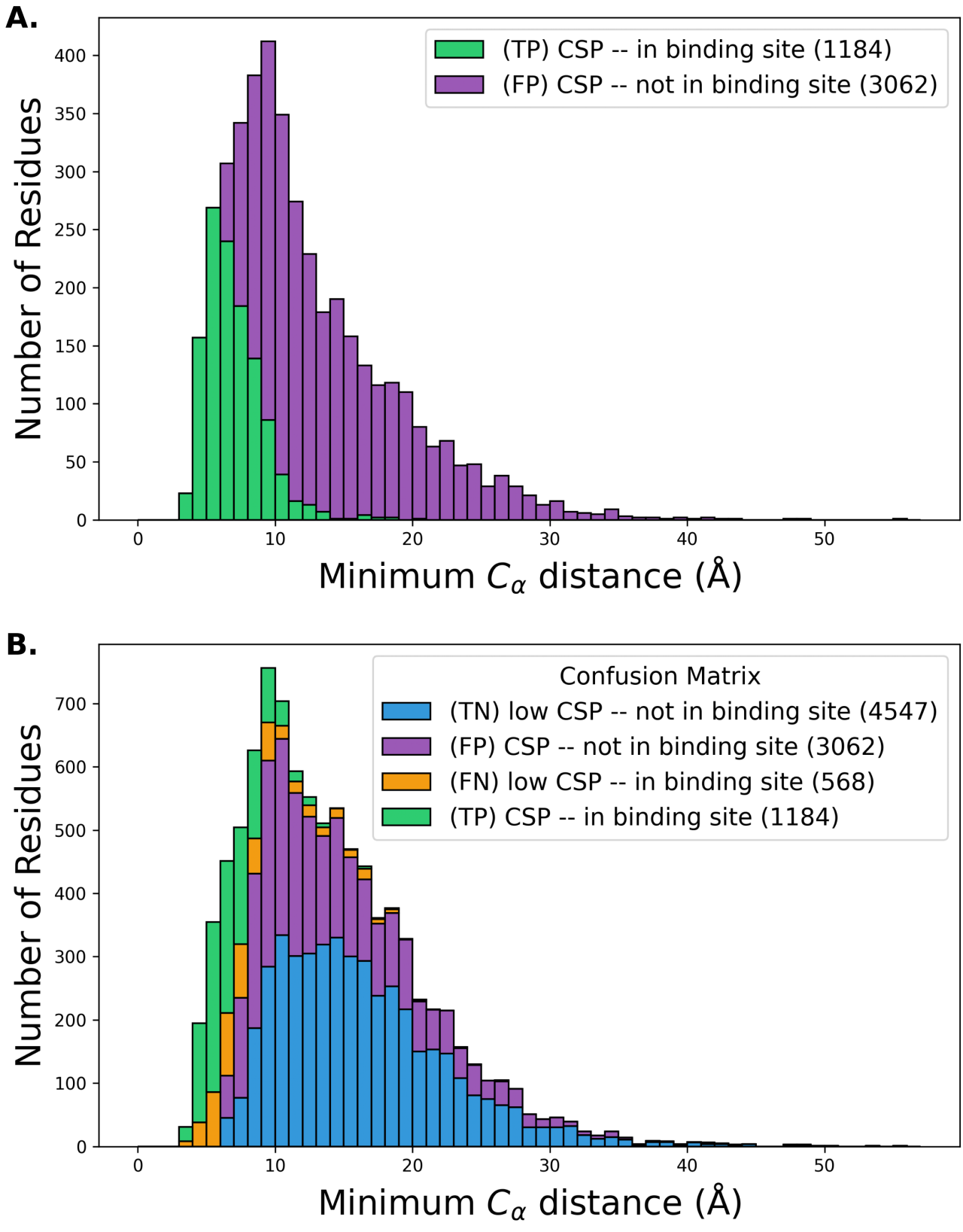


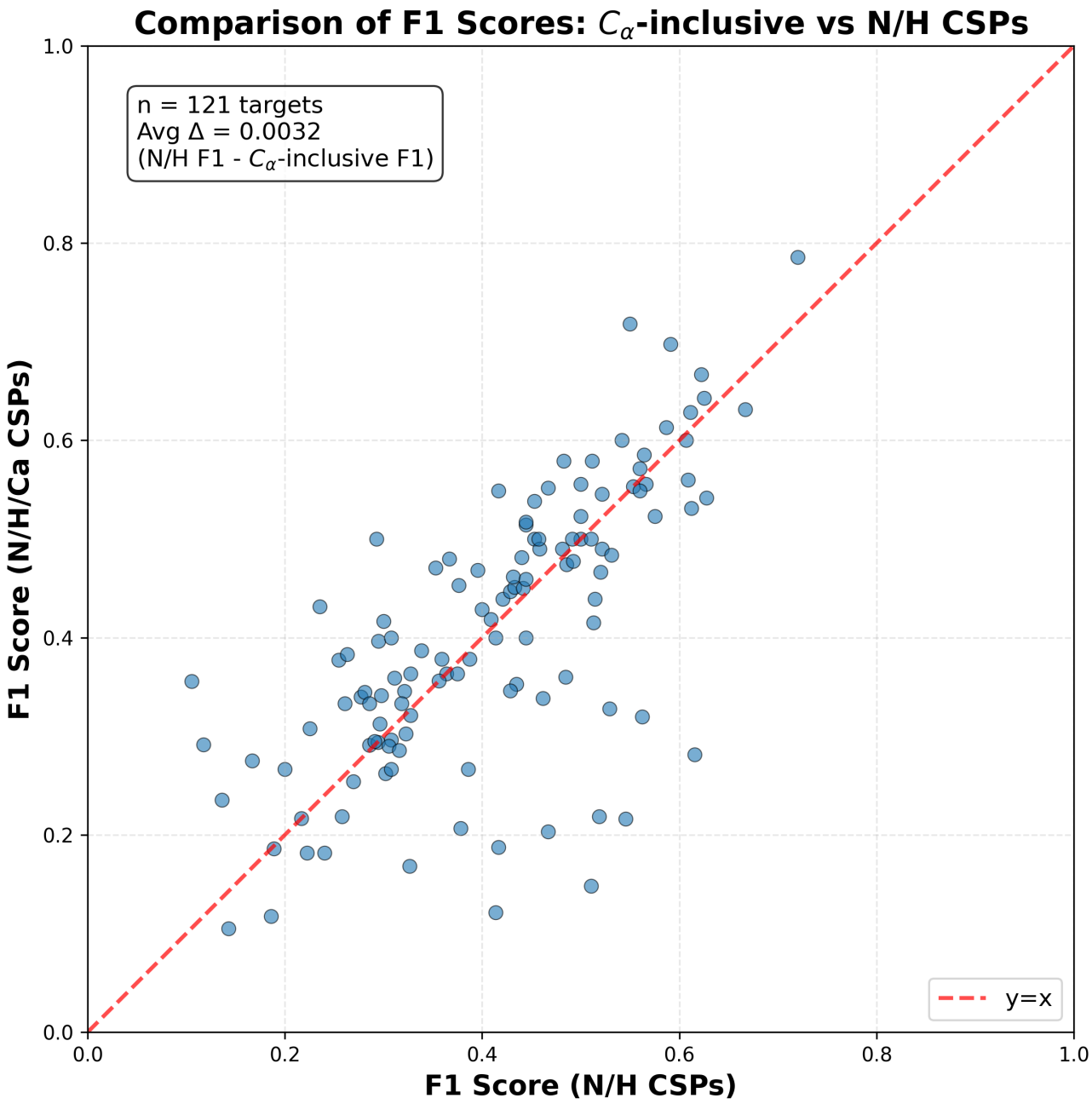
SI Fig. S9. Figure 3 histograms for the same-author/same-study subset. Targets are the curated pairs in `CSP_UBQ_ph0.5_temp5C_same_author_list.csv` (n = 32 with pipeline outputs of 33 curated pairs): buffer-similar apo/holo systems that share a study or last author between apo and holo chemical-shift depositions. **A.** significant-CSP TP vs FP by minimum Cα distance to the interface; **B.** full confusion-matrix stacks (TN/FP/FN/TP). This figure reports results on a very high-fidelity subset, and the results are very similar to those reported in Figure 3: 67.0% of CSPs are distal vs 69.4% of CSPs are distal in the n = 135 subset reported in Figure 3.



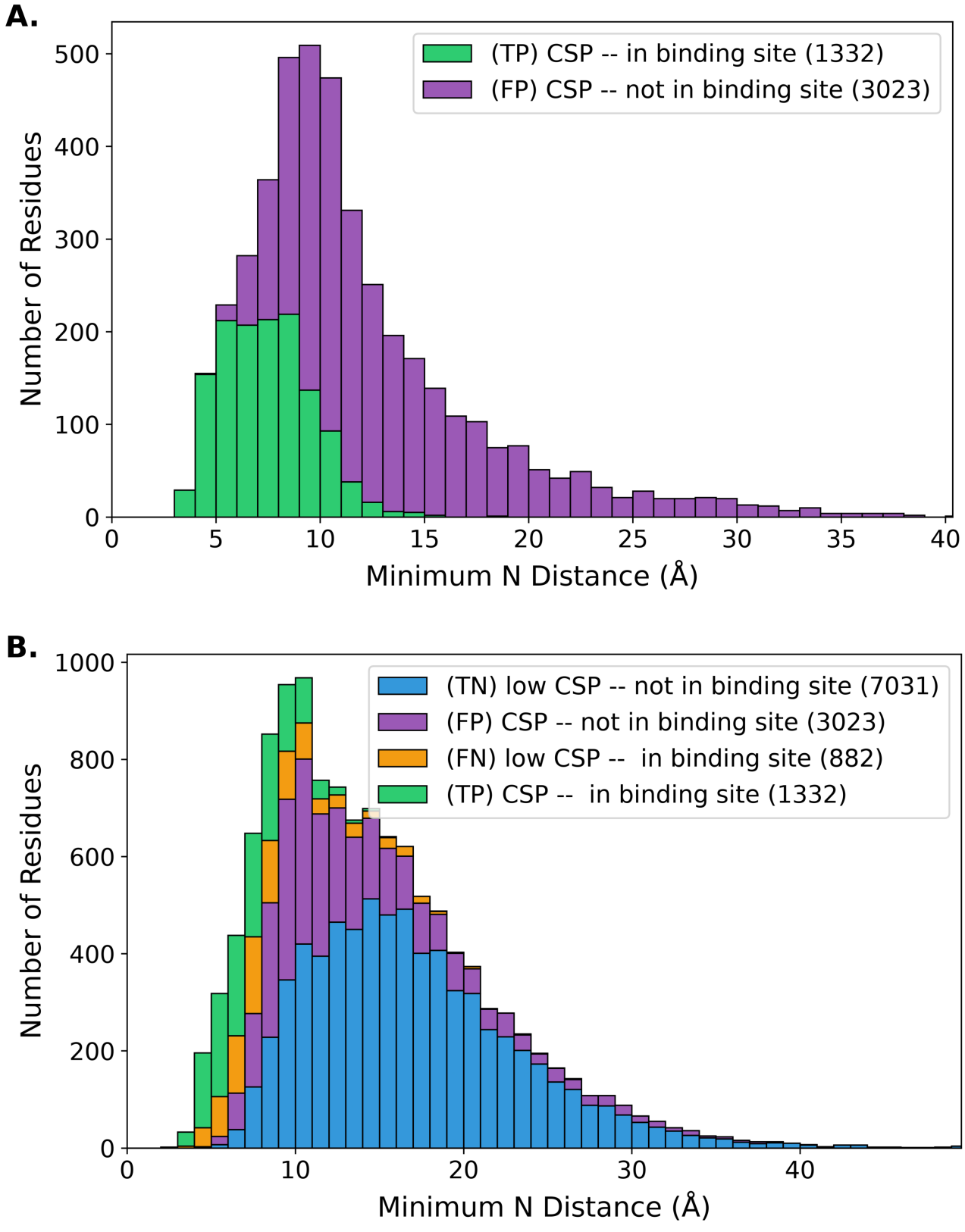
SI Fig. S10. Confusion histograms calculated with Eqn. 2 for the $n = 121$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with CA row coverage $> 50\%$. These results are very similar to CSPs calculated using SI Equation S1. 72.2% of CSPs using SI Equation S2 are distant from the binding site compared to 69.4% using SI Equation S1.



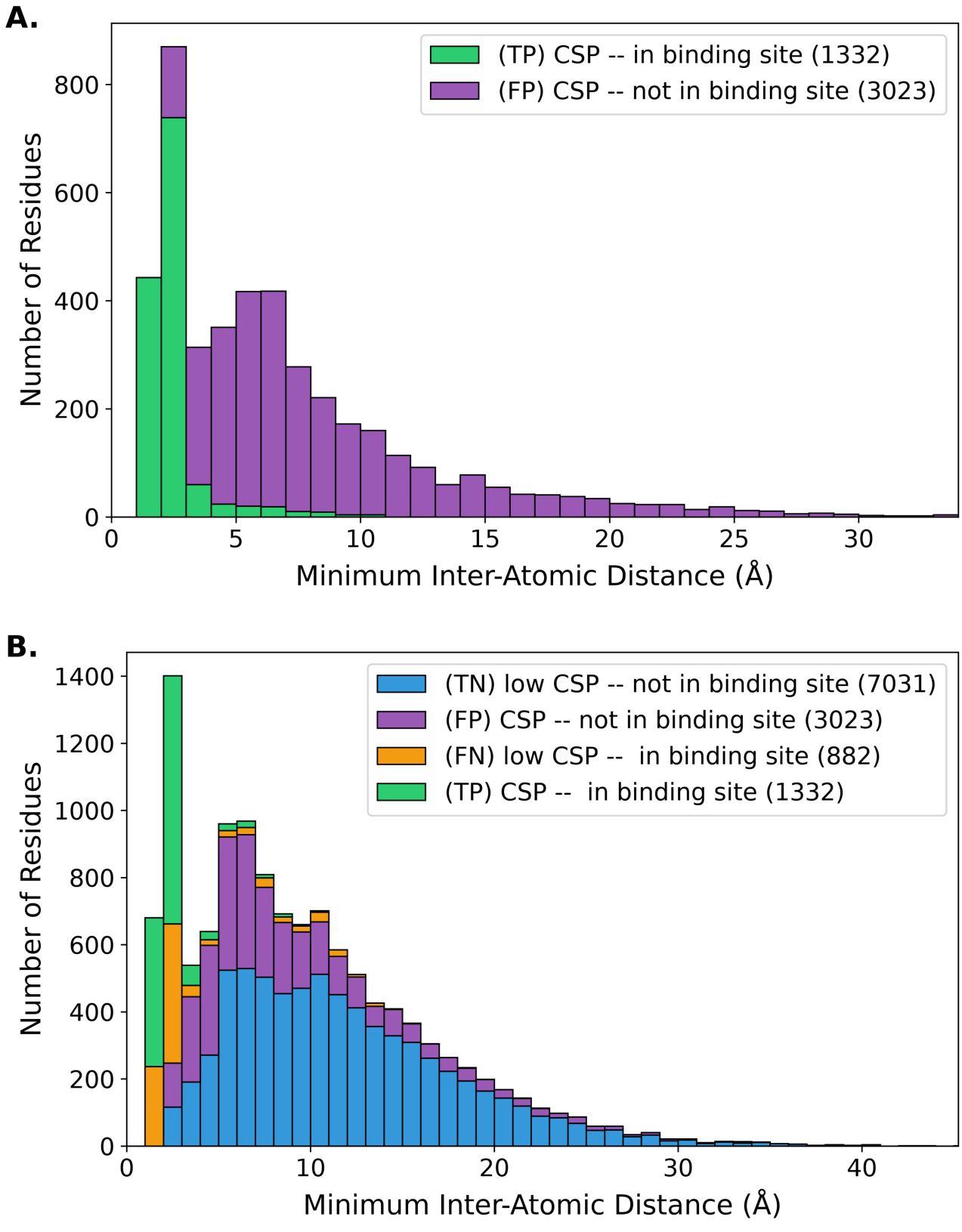
SI Fig. S11. Confusion histograms calculated with Eqn. 3 for the buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` that also have Hα and Cα shifts in both apo and holo experiments ($n = 101$). These results are very similar to CSPs calculated using SI Equation S1. 72.2% of CSPs using SI Equation S3 are distant from the binding site compared to 69.4% using SI Equation S1.



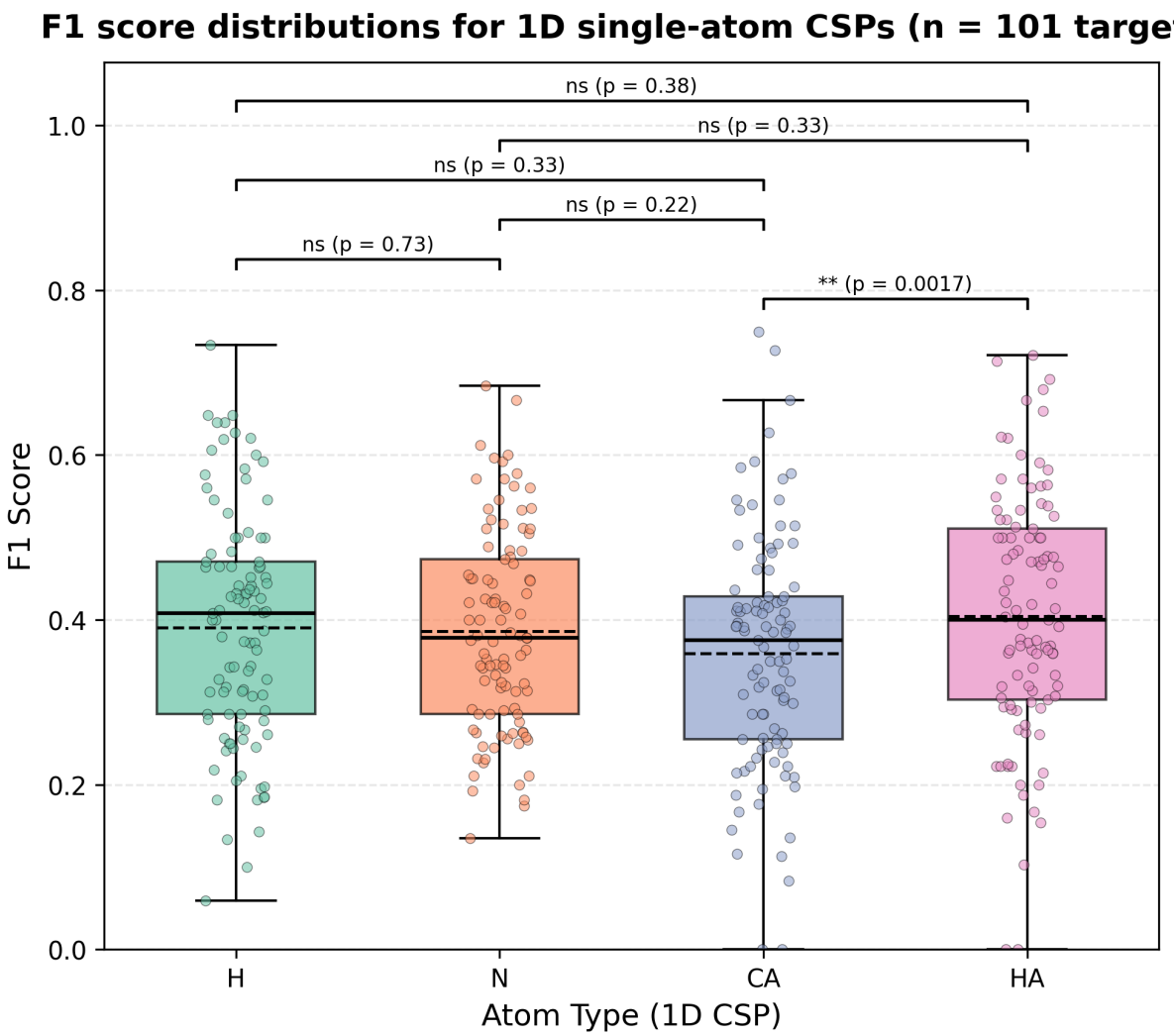
SI Fig. S12. F1 for Eqn. 1 CSPs vs Eqn. 2 CSPs ($n = 121$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with CA row coverage $> 50\%$). F1 scores on the y axis use SI Equation S2 (combined HN+Cα); F1 scores on the x axis use SI Equation S1 (combined HN). The F1 scores here demonstrate that per-target there are very few differences in the distribution of CSPs distant from the binding site across the $n = 121$ subset of targets with CA CSPs and similar experimental conditions.



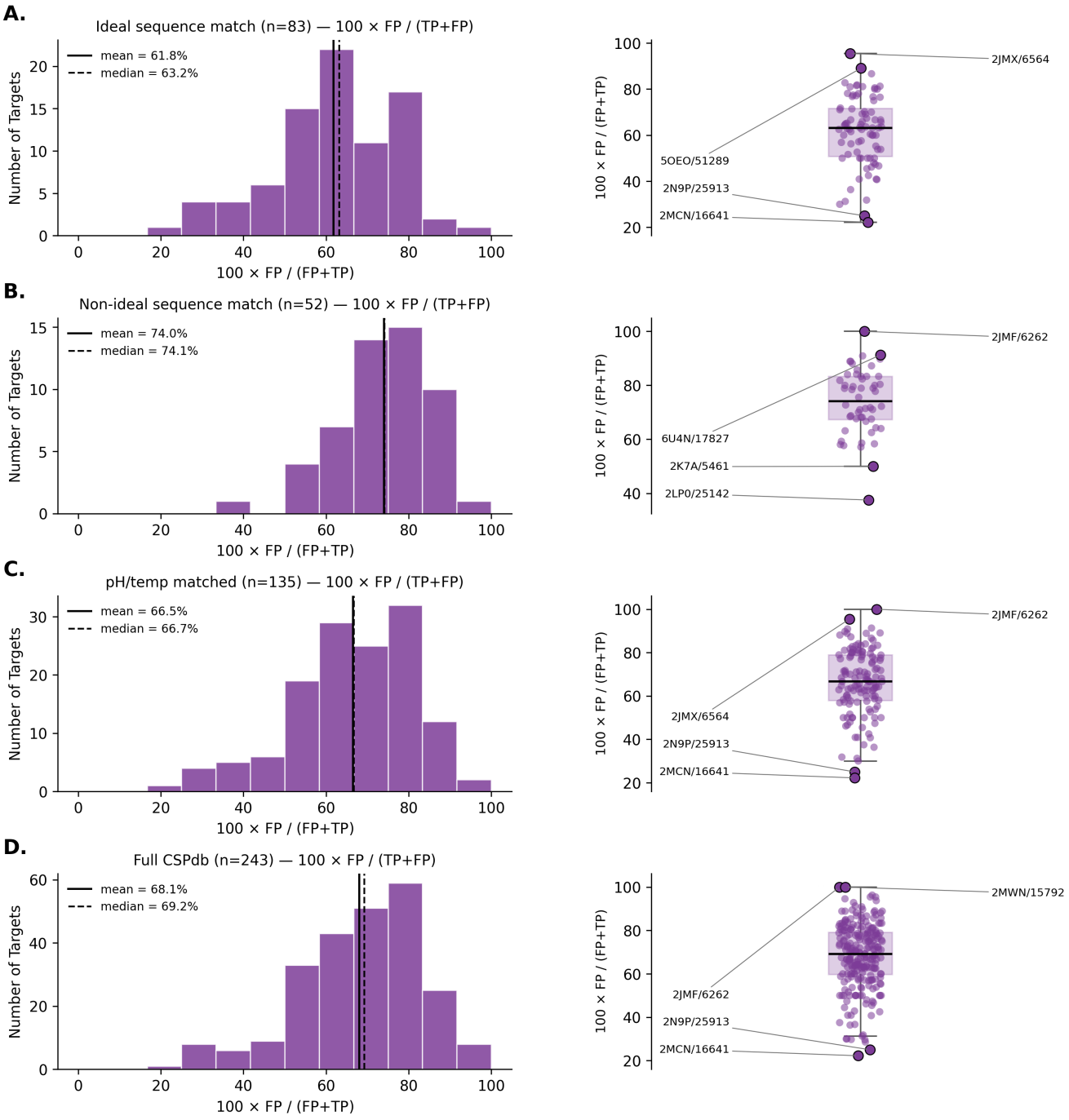
SI Fig. S13. Confusion histograms by closest interchain N–N distance (n = 135 buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs).



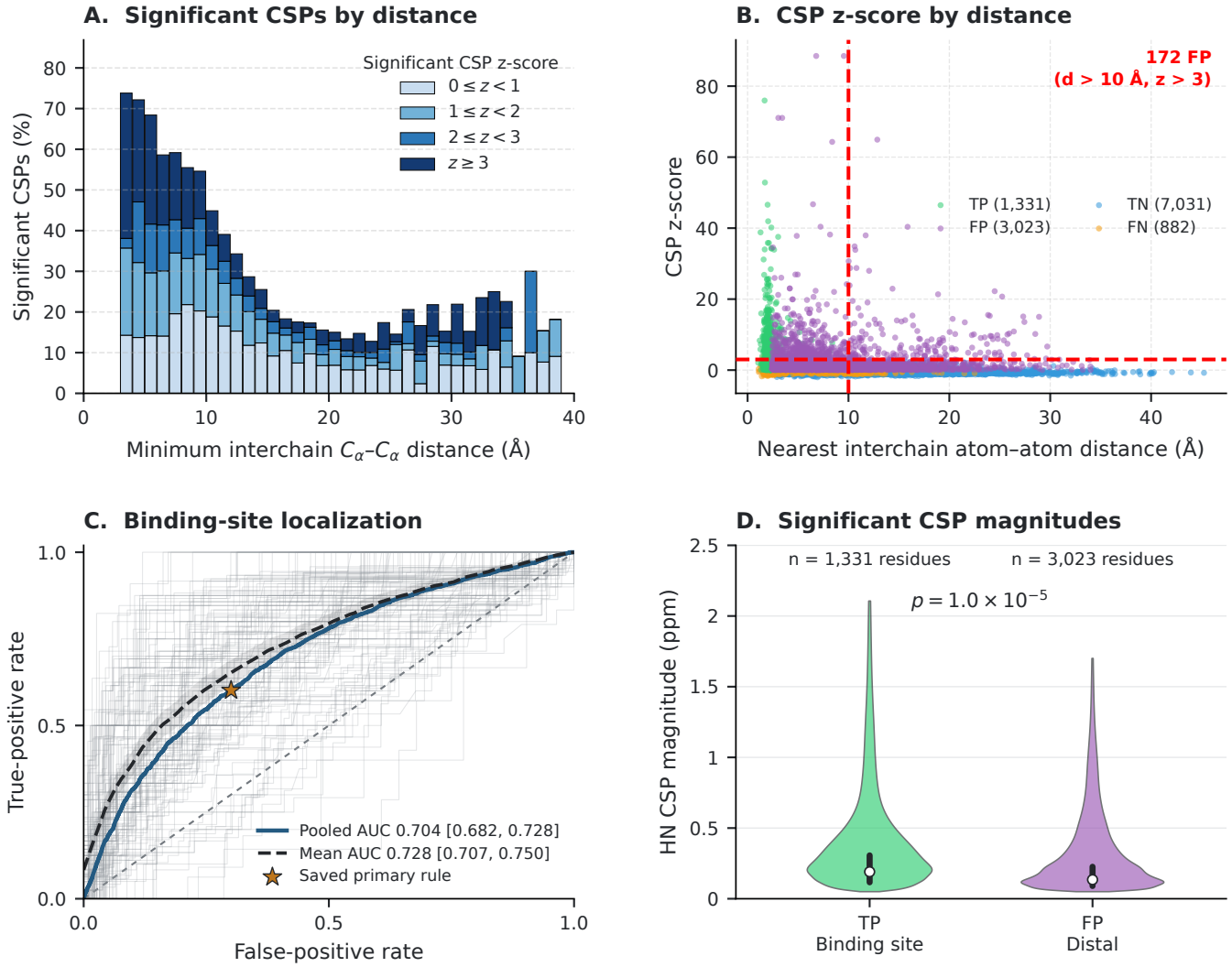
SI Fig. S14. Confusion histograms by closest interchain any-atom distance ($n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs).



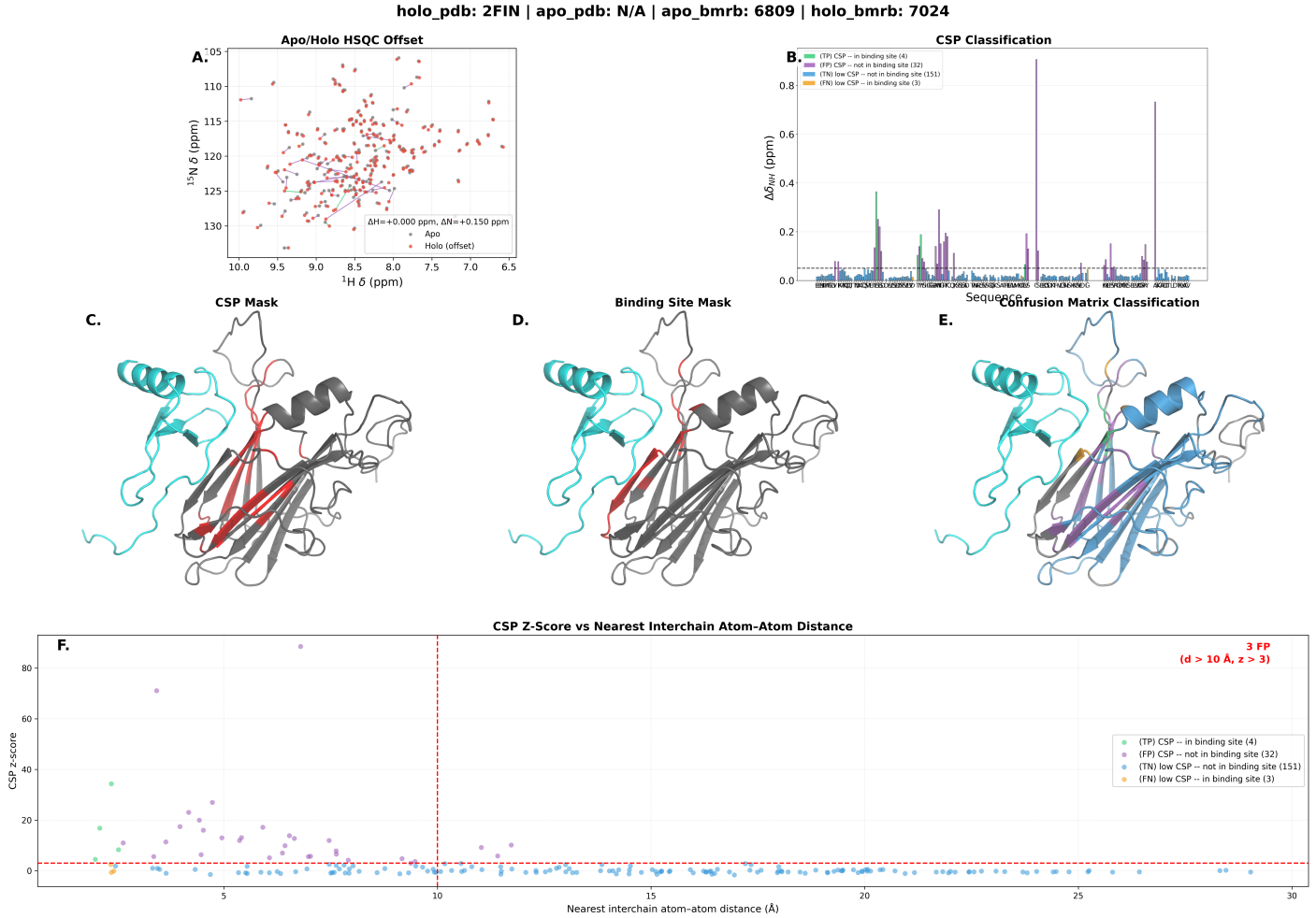
SI Fig. S15. Per-target F1 scores for 1D H/N/Cα/Hα CSPs (n = 101 buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with paired H/N/CA/HA F1 scores). Boxes show the per-target F1 distribution for 1D CSPs of each nucleus. Pairwise comparisons are two-sided Wilcoxon signed-rank tests for all unordered atom pairs, with Holm–Bonferroni adjustment of p-values. After Holm–Bonferroni adjustment, H, N, and Cα F1 scores do not differ (all adjusted $p > 0.05$). Cα versus Hα is the exception (Holm-adjusted $p = 0.0017$): median per-target F1 is 0.38 for Cα and 0.40 for Hα. Distal CSPs still occur for every nucleus at a similar frequency. Apo/holo offsets for 1D CSPs are chosen independently for each nucleus by a 1D grid search on unreferenced shifts, using the same bounds as the joint searches ($^{1}H_N$ $\pm$0.2 ppm, step 0.01 ppm; ^{15}N $\pm$1.5 ppm, step 0.05 ppm; $^{13}C\alpha$ $\pm$2 ppm, step 0.05 ppm; $^{1}H\alpha$ $\pm$0.12 ppm, step 0.01 ppm). The CSPs calculated to retrieve an F1 score per target are calculated as $(\mathrm{dd}\langle\mathrm{Shift}\rangle^2)^{1/2}$ and the threshold for significance is set as the max of the cleaned mean and 0.05 ppm, for all 1D CSPs shown in this figure.



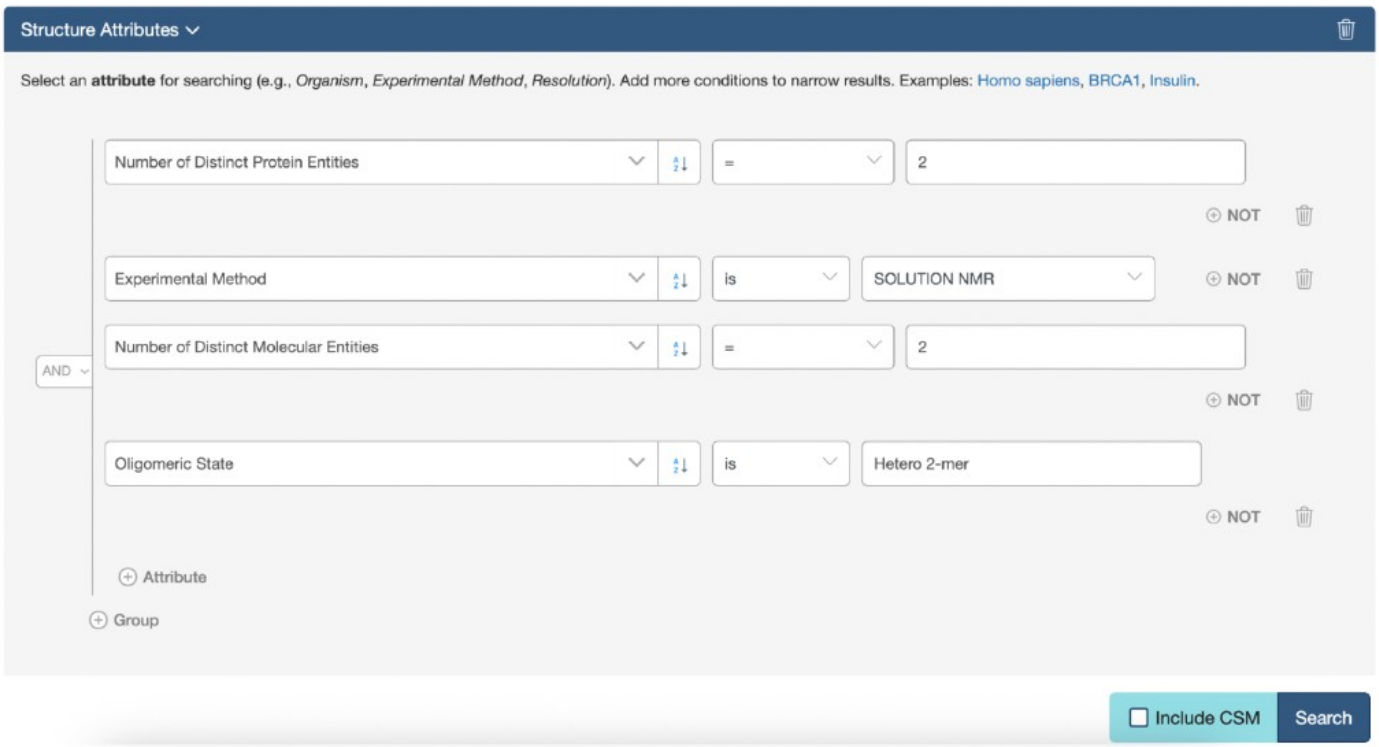
SI Fig. S16. Per-target false-positive rates under the primary CSP cutoff max(cleaned mean, 0.05 ppm). **A.** ideal sequence match ($n = 83$); **B.** non-ideal sequence match ($n = 52$); **C.** pH/temp matched ($n = 135$); **D.** full CSPdb ($n = 243$). For each cohort, panels report $100 \times \mathrm{FP}/(\mathrm{TP} + \mathrm{FP})$ among significant CSPs. This figure shows per-target false positive rates across four subsets, and reveals there are minimal differences between these four subsets.



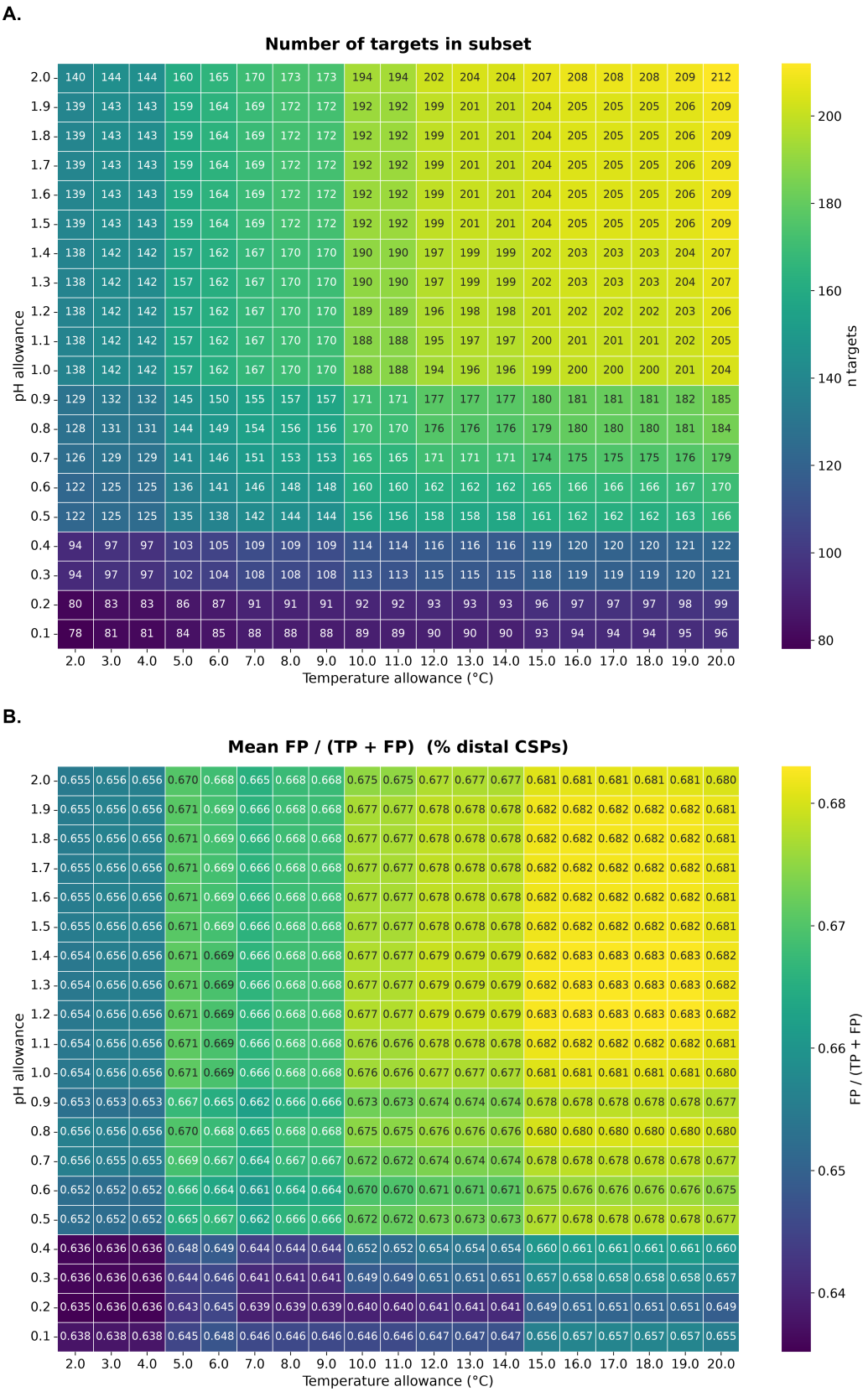
SI Fig. S17. CSPs and structural binding-site localization. All panels use 135 pH- and temperature-matched apo/holo targets. The per-target cutoff is max(cleaned mean HN CSP, 0.05 ppm). **A.** Percentage of residues with a significant HN CSP among those with a recorded CSP in each 1 Å bin of minimum interchain C_α–C_α distance. Stacks partition the significant fraction by CSP z-score. **B.** Per-residue CSP z-score versus nearest interchain atom–atom distance. For residue i, $z_i = (\mathrm{CSP}_i - \mu_{\mathrm{clean}})/\sigma_{\mathrm{clean}}$, where μ_{clean} and σ_{clean} are the mean and population standard deviation of the target's HN CSP list after iterative removal of high outliers. Colors indicate true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) relative to the structural binding-site definition. Dashed lines mark $z = 3$ and $d = 10$ Å. The 172 annotated FPs exceed both thresholds. **C.** Receiver operating characteristic (ROC) curves across HN CSP thresholds for 12,267 eligible residues. Gray curves show individual targets, blue shows the pooled ROC using CSP divided by the target-specific cutoff, and the black dashed curve shows the equal-target mean ROC. The pooled area under the curve (AUC) is 0.704 (95% CI 0.682–0.728), and the mean per-target AUC is 0.728 (95% CI 0.707–0.750). The band shows the pointwise 95% interval for the mean ROC. Intervals use 2,000 whole-target bootstrap replicates. The gray diagonal indicates chance, and the star marks the pooled operating point from the saved primary significance calls. **D.** HN CSP magnitudes for 1,331 significant binding-site residues (TP, green) and 3,023 significant distal residues (FP, purple) under the saved primary rule, pooled across targets. Scott-bandwidth kernel densities are estimated and normalized to equal area in $\log_{10}$ CSP space and bounded by the observed ranges. White dots mark medians and black bars mark interquartile ranges. The y-axis is linear (ppm). The symmetric Kullback–Leibler divergence was 0.0941 nats (20 shared log-spaced bins, 0.5 pseudocount per bin), with $p = 1.0 \times 10^{-5}$ from 99,999 within-target label permutations.



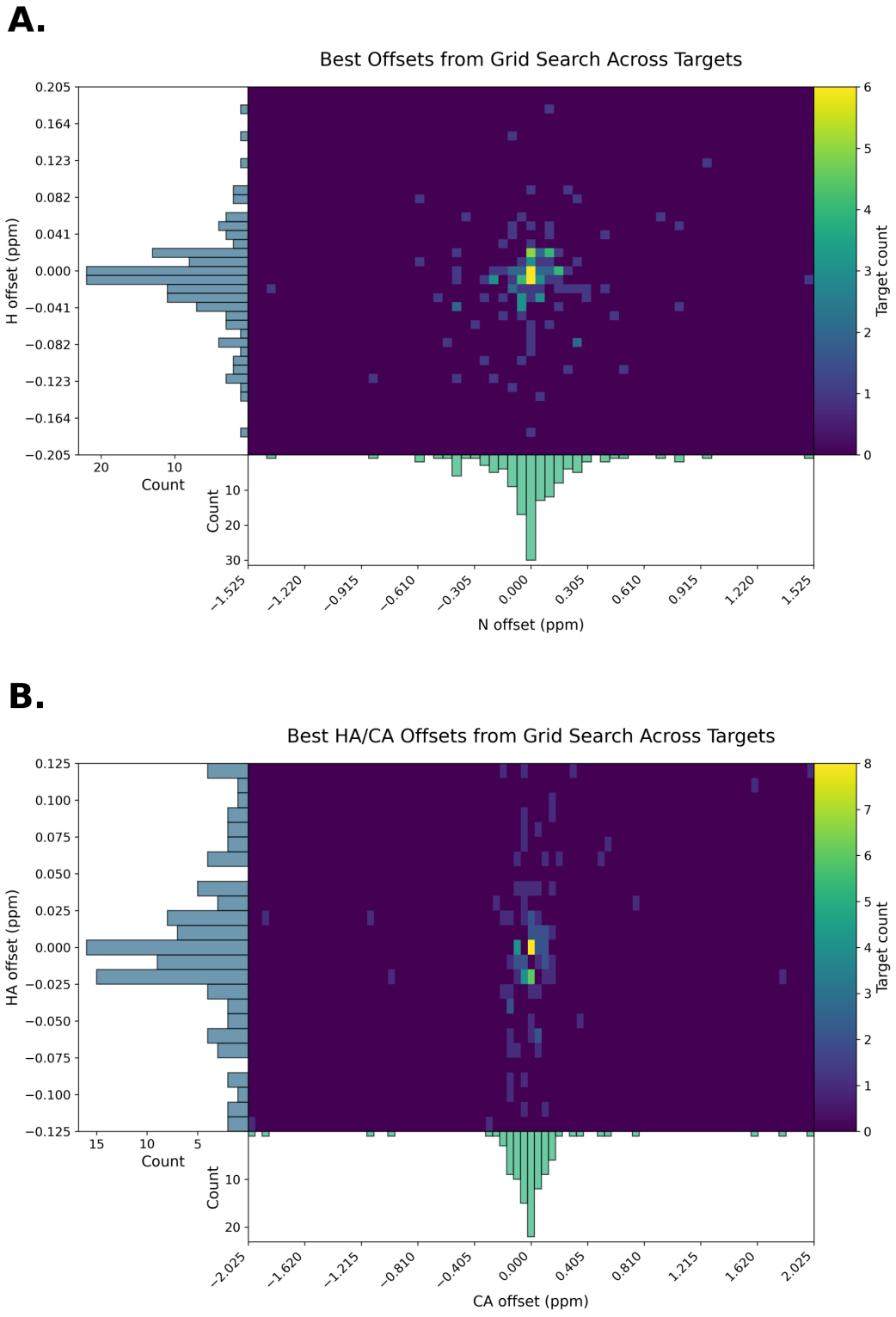
SI Fig. S18. Case study for holo 2FIN / apo BMRB 6809 (holo BMRB 7024). **A.** ^{15}N/^{1}H HSQC scatterplot overlaying the apo and holo assigned spectra with the optimal N/H offset applied. **B.** Calculated CSPs for receptor residues along the sequence, bars colored by confusion matrix classification. **C.** The mask of significant CSPs. **D.** The mask of binding-site residues. **E.** The confusion matrix classification projected onto the structure: distal CSPs are purple (FP), CSPs in the binding site are green (TP), non-CSPs in the binding site are orange (FN), and non-CSPs outside of the binding site are blue (TN). Residues without CSP measurements are shown in gray; ligands are colored cyan. **F.** CSP z versus nearest interchain atom–atom distance. Dashed lines mark $z = 3$ and $d = 10$ Å; the annotation counts false positives with $d > 10$ Å and $z > 3$. This plot shows monotonic decay of CSPs with increasing nearest interchain atom–atom distance. Residues in a set of β sheets further away from the ligand with significant shifts have lower CSPs than those closer to the binding site.



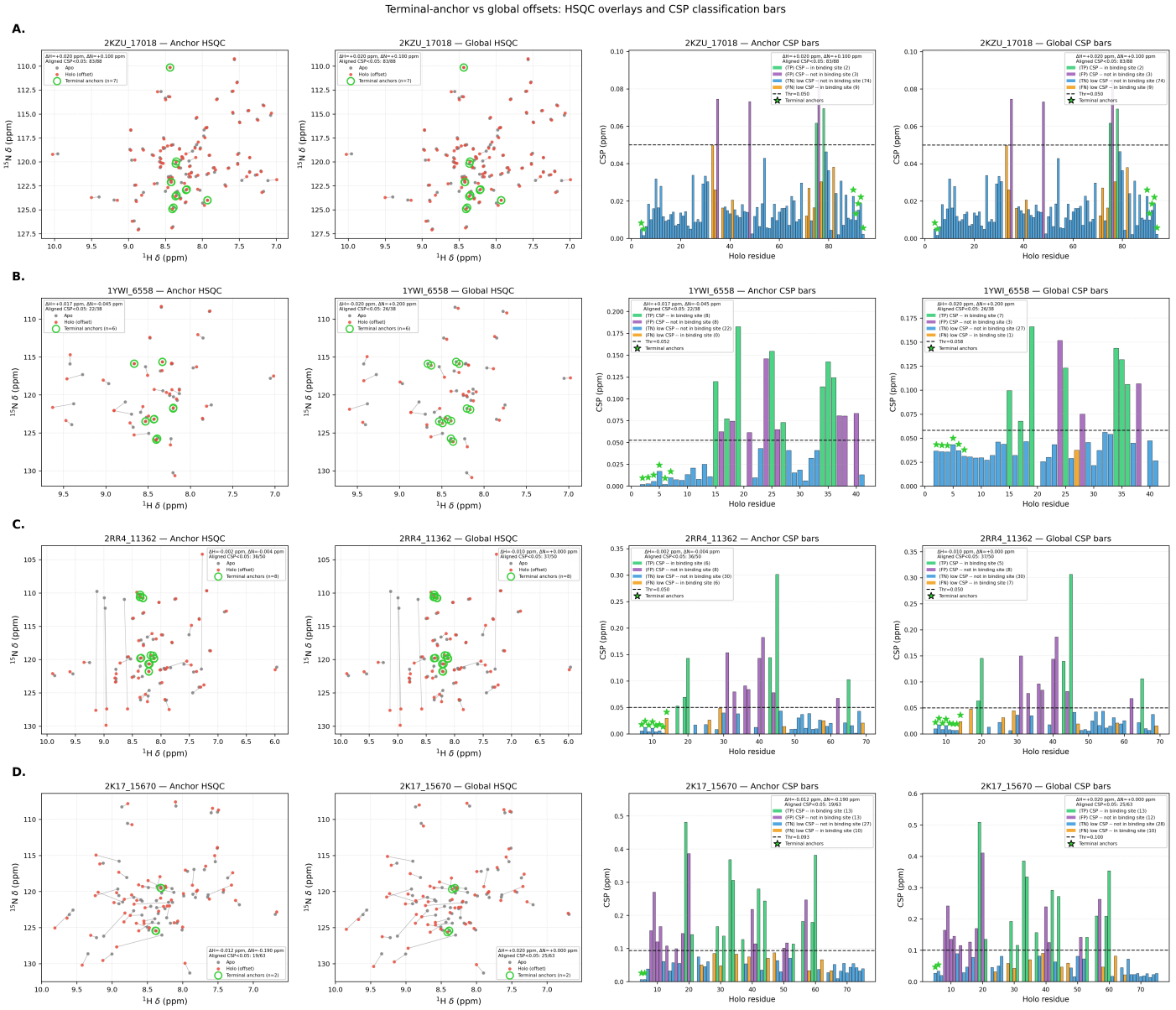
SI Fig. S19. PDB Advanced Search Settings. For the subset of targets manually scraped from the PDB, we restrict our analysis to these settings: entries resolved by Solution NMR, with two and only two distinct protein chains. At the time of curation, this advanced search resulted in 720 hits on the RCSB PDB. From this list of 720, entries were included in the CSP DB if there were N and H chemical shifts deposited in the BMRB for an identical sequence which was evaluated in both apo and holo experiments.



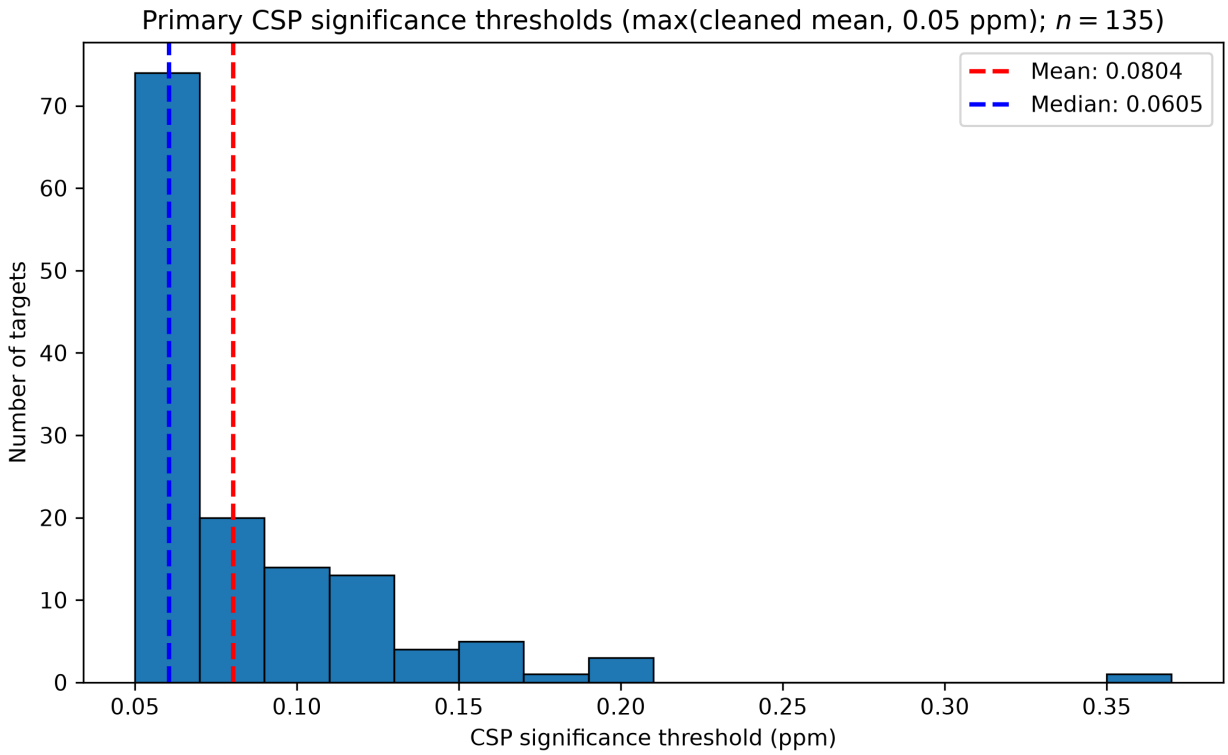
SI Fig. S20. Buffer pH/temperature threshold sweep. A. Number of apo/holo pairs retained as $|\Delta\text{pH}|$ and $|\Delta T|$ thresholds are tightened. **B.** Mean FP/(TP+FP) (percent distal CSPs) in the same subsets. This figure reveals the mean per-target false positive rate changes only slightly across all thresholds in apo/holo experimental condition differences from a minimum of 63.6% to a maximum of 68.3%.



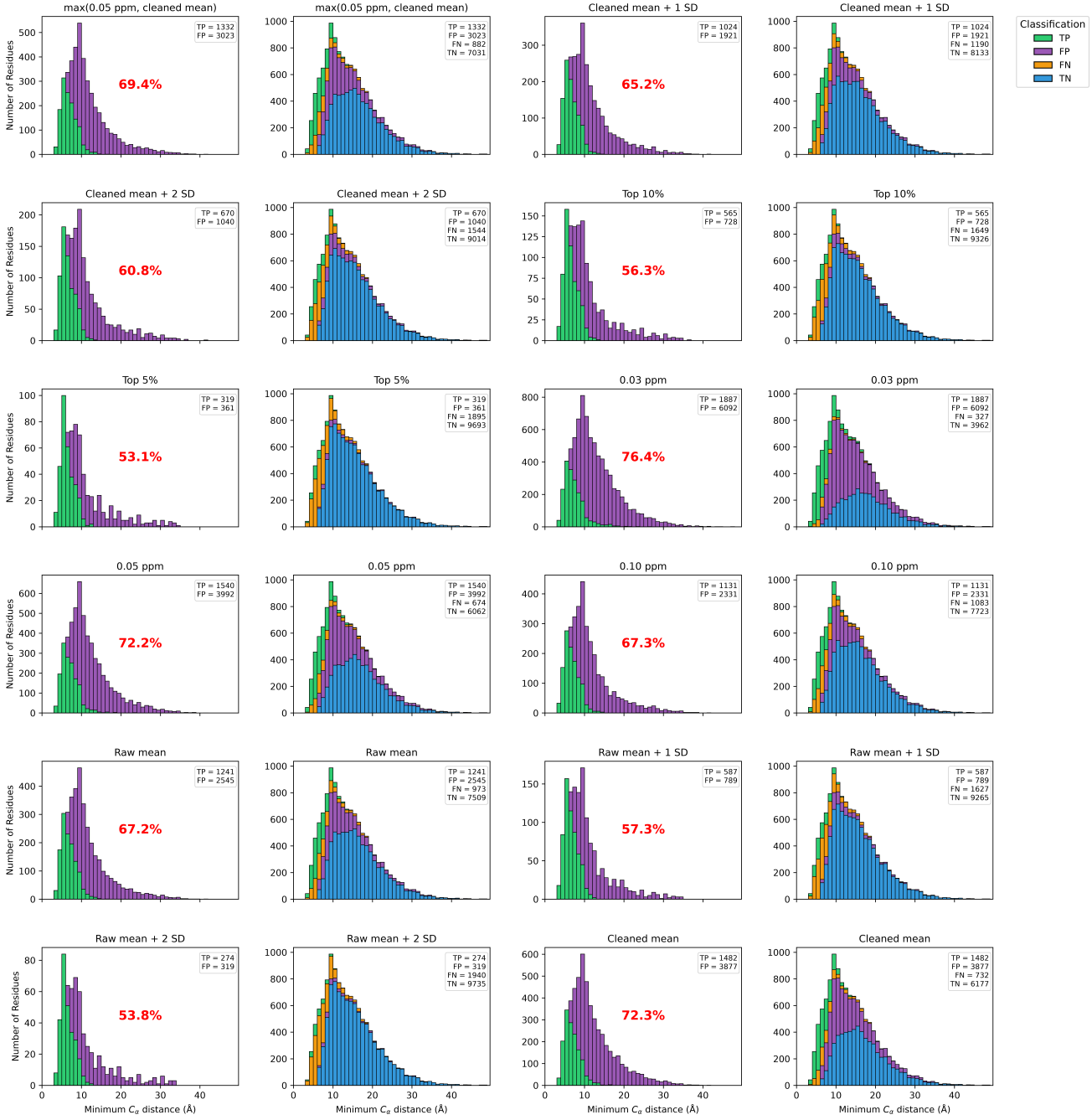
SI Fig. S21. Ideal chemical-shift offset grids for buffer-similar systems in `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs. **A.** H/N referencing ($n = 135$): a grid search selects the H/N offsets that maximize aligned HN peak counts under the pipeline CSP criterion; the heatmap aggregates those optimal ($\Delta\delta_N$, $\Delta\delta_H$) pairs (color = number of targets). Marginal histograms show the one-dimensional distributions of best H (left) and N (bottom) offsets. **B.** HA/CA referencing ($n = 101$ systems with HA/CA grid-search CSVs): the same layout for offsets that maximize aligned HA/CA peak counts, with HA (left) and CA (bottom) marginals. Both nuclei pairs are peaked near zero. On the Cα window of ± 2 ppm, 87 of 101 Cα offsets lie within 0.2 ppm of zero and 5 have $|\text{offset}| \geq 1.5$ ppm, so most apo/holo pairs still require only small referencing corrections.



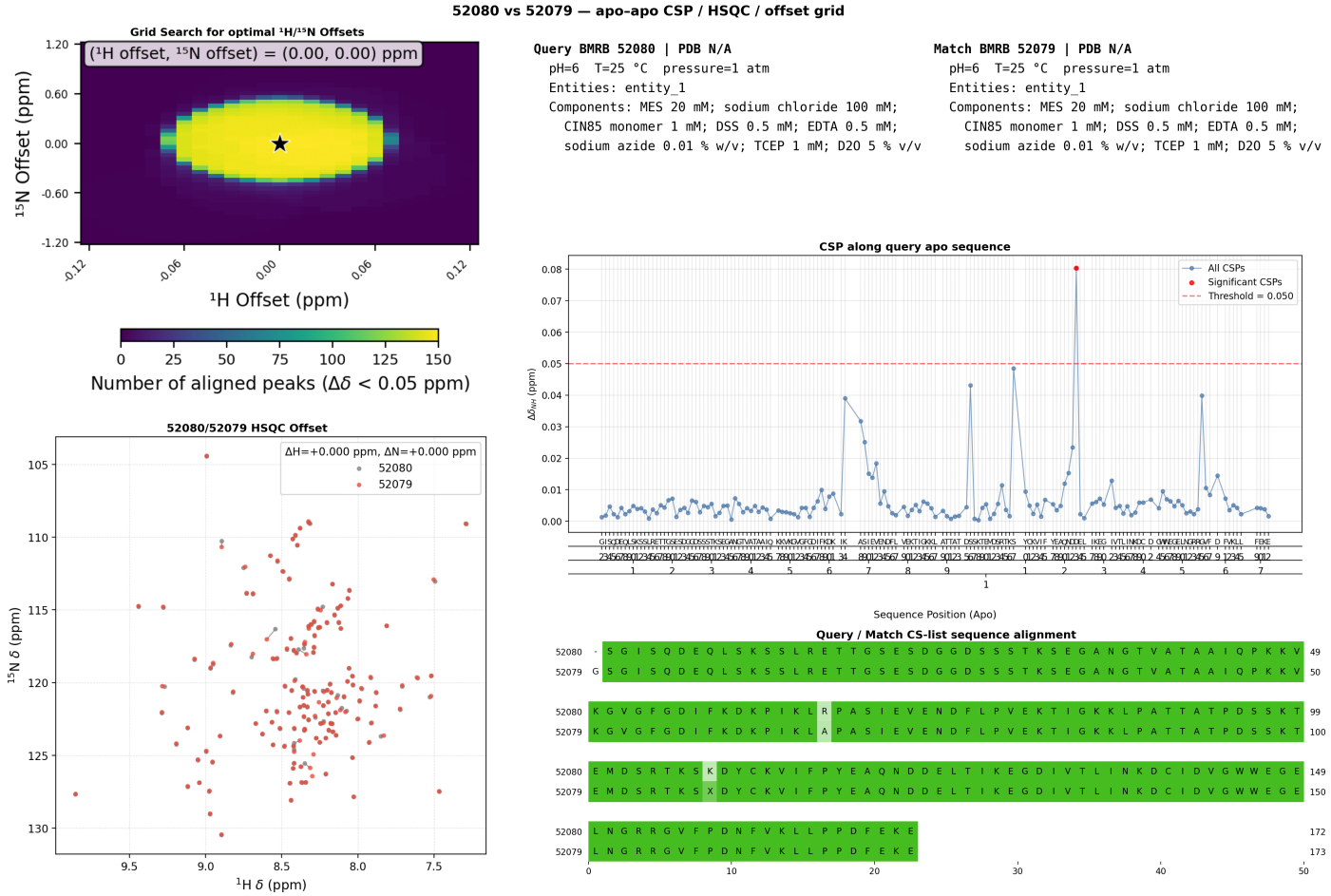
SI Fig. S22. Terminal-anchor vs global offsets: HSQC overlays and CSP classification bars for four targets (2KZU/17018, 1YWI/6658, 2RR4/11361, 2K17/15670). For each target (rows A.–D.): left pair shows apo (grey) vs offset holo (red) HSQC overlays using terminal-anchor RMSD offsets vs the global CSP-count grid; right pair shows corresponding CSP classification bars (TP/FP/TN/FN) under the primary significance threshold. Green markers denote terminal-anchor residues. This figure reveals an alternative approach to referencing apo and holo shift lists would be an appropriate solution to the problem, but is not generalizeable across the dataset. These results further demonstrate our approach of global optimization described in the Methods is an appropriate generalizeable solution to this problem.



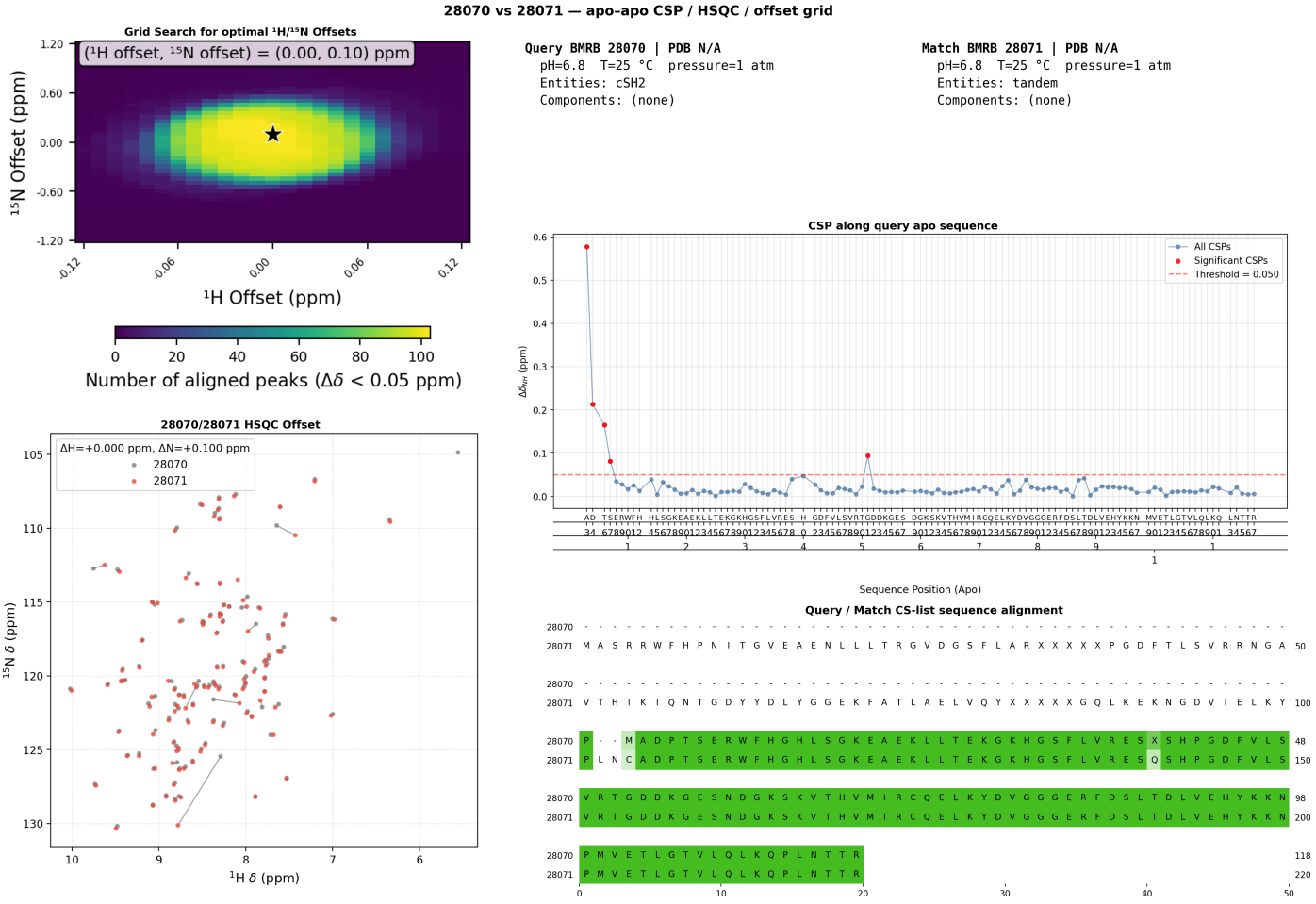
SI Fig. S23. Distribution of CSP significance thresholds (primary cutoff max(cleaned mean, 0.05 ppm); $n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs). Mean = 0.0804 ppm. This figure shows our mean CSP threshold is 0.08 ppm, corresponding to ∼50 Hz at 600 MHz, which is 2–5× larger than the corresponding proton resonance linewidths.



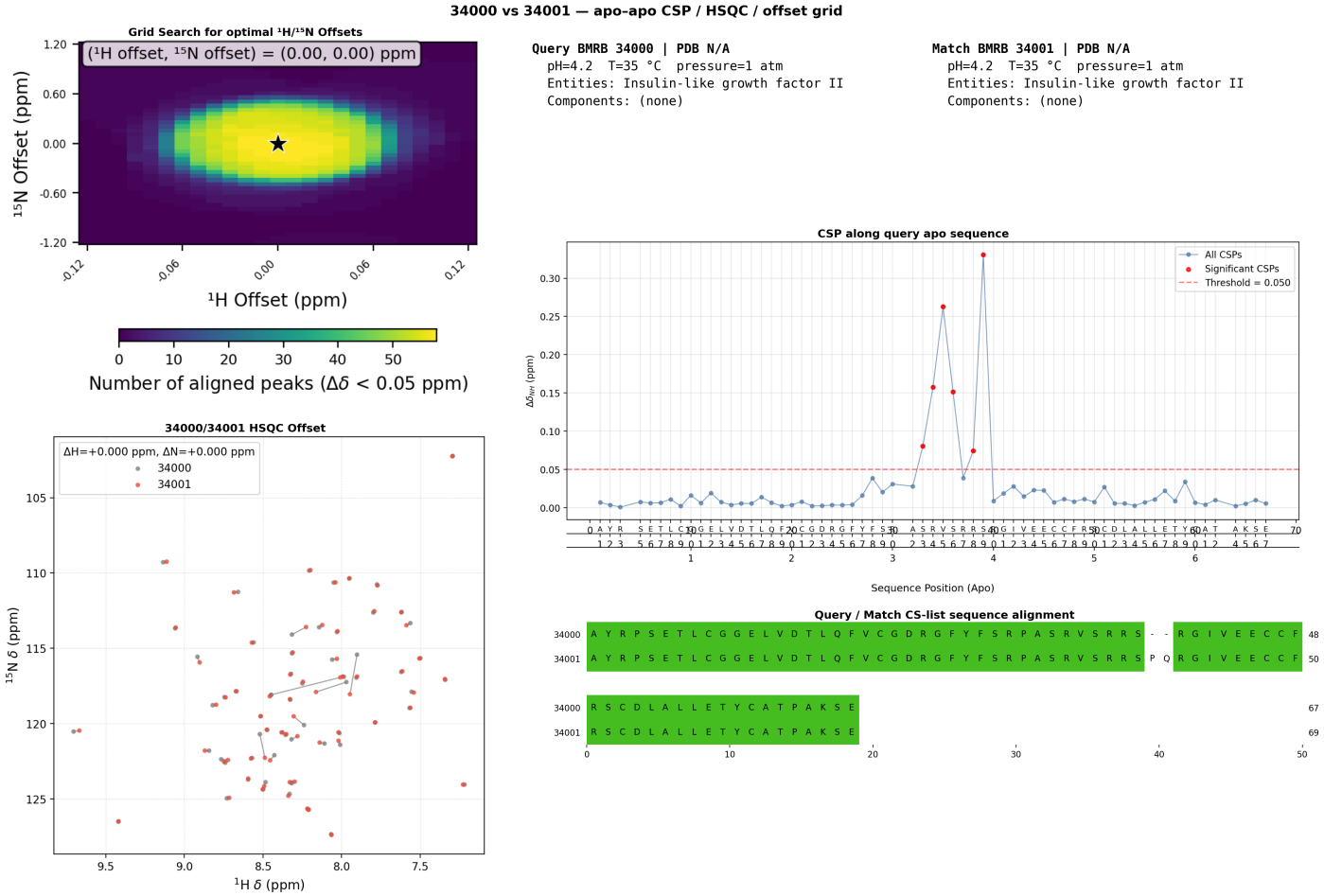
SI Fig. S24. Figure 3-style CA-distance histograms under alternate HN CSP significance thresholds ($n = 135$ buffer-similar systems from `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs). Each column is a 3a (TP/FP among significant CSPs) | 3b (full confusion) pair. Thresholds (left to right, then continuing as labeled on the figure): **primary** max(cleaned mean, 0.05 ppm) after iterative outlier removal; **cleaned mean** +1 **SD** and +2 **SD**; **top 10%** and **top 5%** of HN CSP values; fixed cutoffs **0.03**, **0.05**, and **0.10 ppm**; **raw mean**, **raw mean** +1 **SD**, and **raw mean** +2 **SD** (no outlier removal); and **cleaned mean** without the 0.05 ppm floor. This figure reveals our results are relatively stable if we use other reasonable thresholds of what is considered a significant CSP. Some thresholds from this figure we determine are too high (e.g. Raw mean + 1 SD, Raw mean + 2 SD, Top 10%, Top 5%) which results in less distal CSPs than is expected from the other thresholding approaches. SI Fig. S17C shows the continuous ROC analysis, and SI Fig. S17D shows the significant-CSP magnitude distributions.



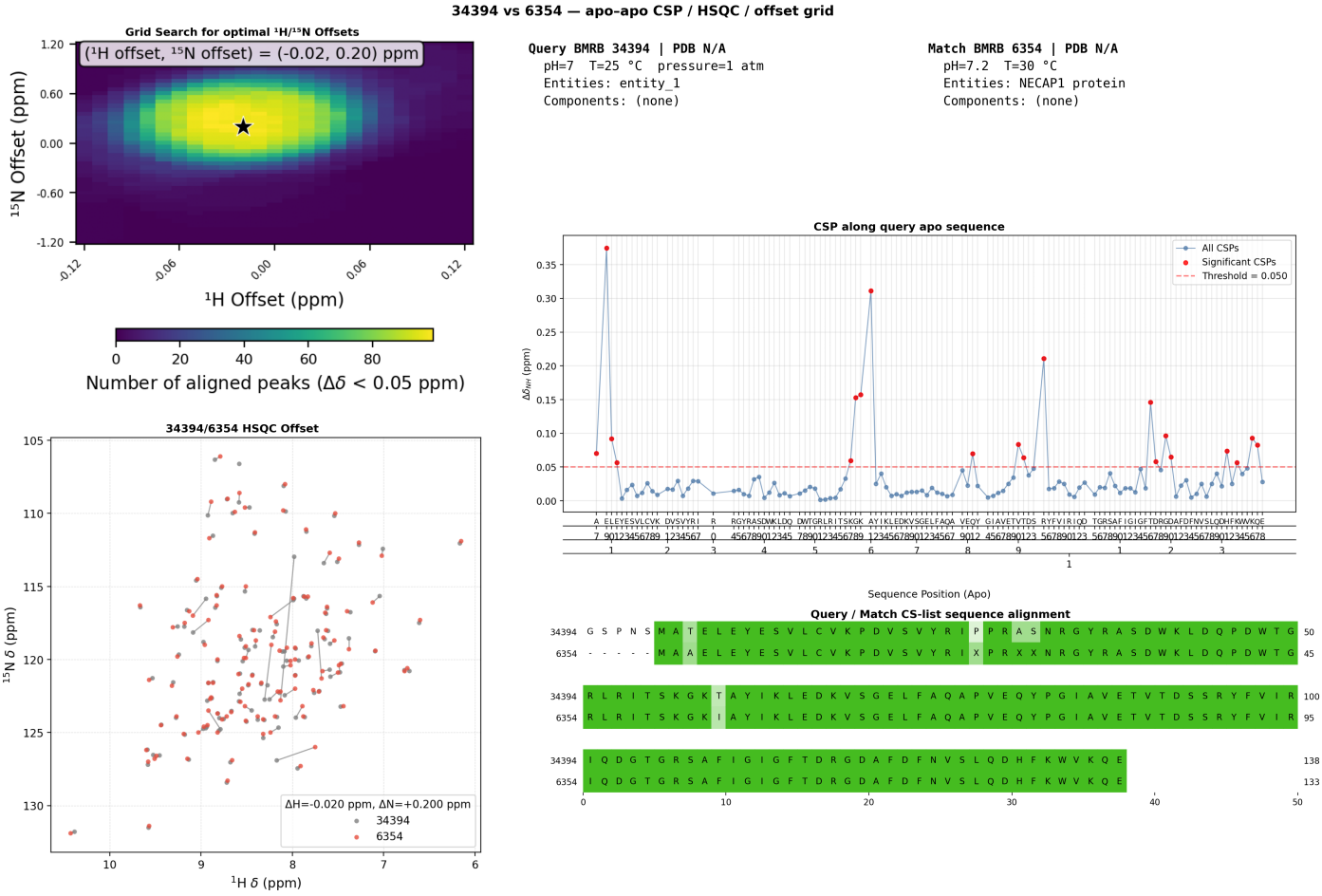
SI Fig. S25A. Apo vs apo control study for BMRB 52080 and 52079. This case involves the CIN85 SH3B–SH3C linker (residues 163–333). There is a sequence mismatch at R227A (BMRB 52080 is the R229A variant; BMRB 52079 is R227A/R229A); BMRB 52079 also has an extra N-terminal Gly. 99.3% of the CSPs for this control are below 0.05 ppm.



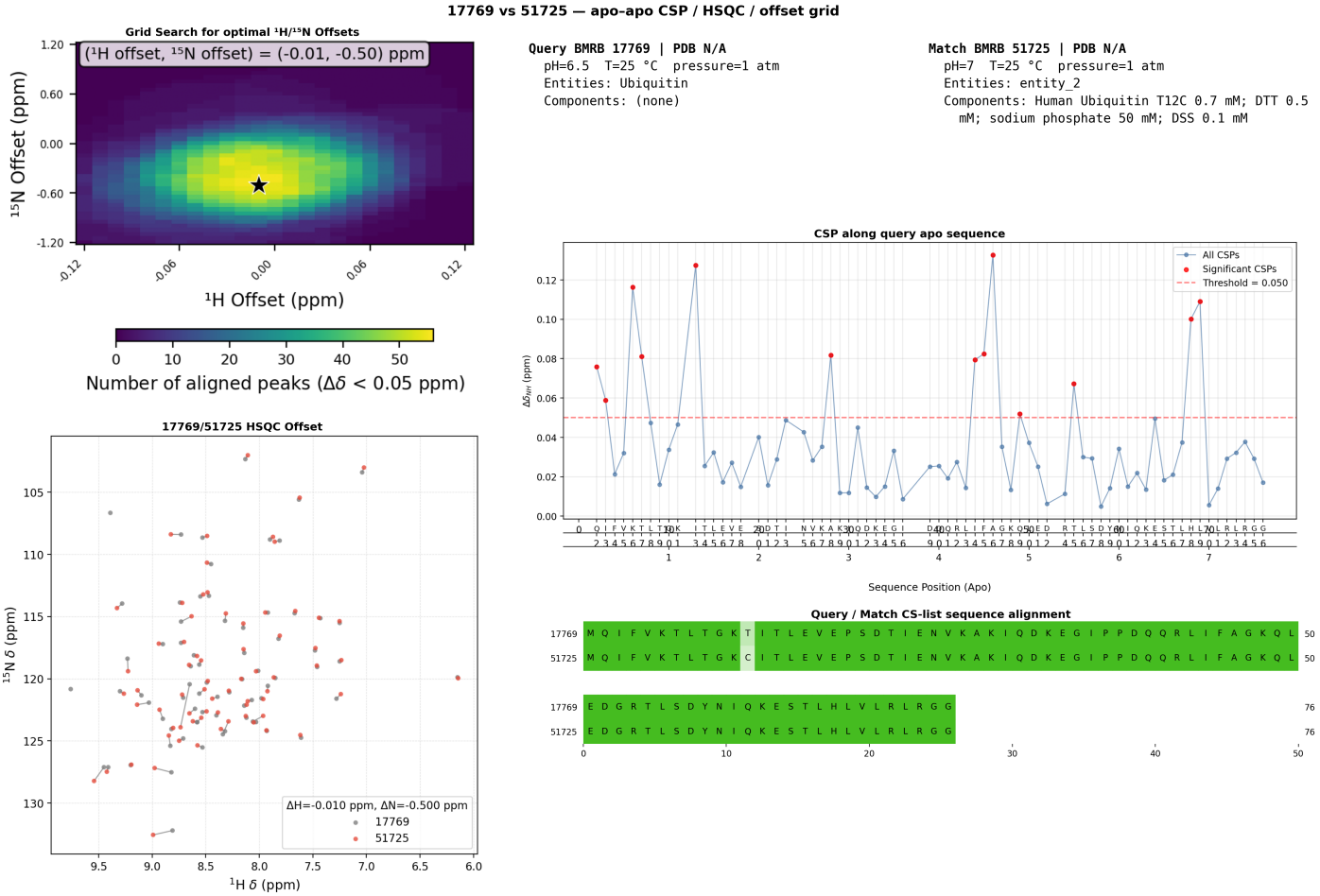
SI Fig. S25B. Apo vs apo control study for BMRB 28070 and 28071. This case involves the SH2 domains of SHP2. BMRB 28071 is the tandem nSH2–cSH2 construct and includes a large N-terminal region (the nSH2 domain and inter-SH2 linker, ∼100 residues) that is absent from the isolated cSH2 entry (BMRB 28070). 95.4% of the CSPs for this control are below 0.05 ppm.



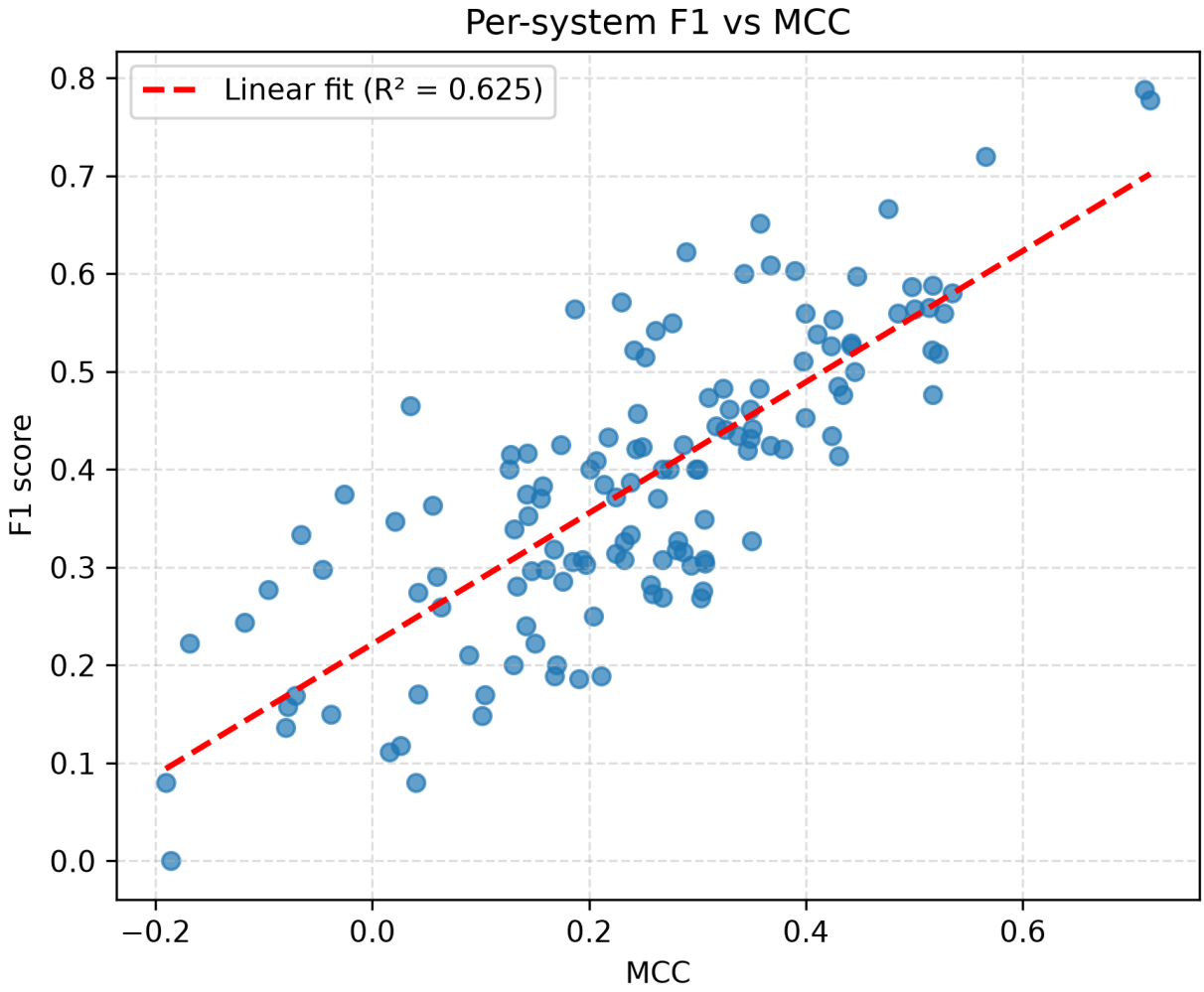
SI Fig. S25C. Apo vs apo control study for BMRB 34000 and 34001. This case involves insulin-like growth factor II (IGF-II). BMRB 34001 is the [S39_PQ]-IGF-II variant and contains a two-residue Pro–Gln insertion after Ser39 relative to wild-type IGF-II (BMRB 34000). 90.6% of the CSPs for this control are below 0.05 ppm.



SI Fig. S25D. Apo vs apo control study for BMRB 34394 and 6354. This case involves the NECAP1 PHear domain. BMRB 34394 has a five-residue N-terminal extension (GSPNS) relative to BMRB 6354, plus substitutions T8A and K59I and short internal insertions (Pro28 and Ala31–Ser32) in the 34394 construct. 83.2% of the CSPs for this control are below 0.05 ppm.



SI Fig. S25E. Apo vs apo control study for BMRB 17769 and 51725. This case involves ubiquitin. There is a sequence mismatch at position 12 (T12C; wild-type Thr in BMRB 17769, Cys in BMRB 51725). 81.2% of the CSPs for this control are below 0.05 ppm.



SI Fig. S26. F1 vs MCC across the $n = 135$ buffer-similar systems in `CSP_UBQ_ph0.5_temp5C.csv` with pipeline outputs. This figure reveals a strong correlation between F1 and MCC measures against the $n = 135$ subset. F1 and MCC are both useful metrics to gauge at a glance the extent of distal CSPs are present in the receptor-ligand system[11].

$$\mathrm{CSP} = \sqrt{\tfrac{1}{2}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2\right)}$$

SI Equation S1. Combined HN chemical-shift perturbation

$$\mathrm{CSP} = \sqrt{\tfrac{1}{3}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2 + (0.3\,\Delta\delta_{\mathrm{C}\alpha})^2\right)}$$

SI Equation S2. Combined HN+Cα chemical-shift perturbation

$$\mathrm{CSP} = \sqrt{\tfrac{1}{2}\left(\Delta\delta_{\mathrm{H}\alpha}^2 + (0.3\,\Delta\delta_{\mathrm{C}\alpha})^2\right)}$$

SI Equation S3. Combined Hα+Cα chemical-shift perturbation

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